SATWICK SHAW

DATA SCIENCE & AI ENGINEER | FULL-STACK DEVELOPER

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SUMMARY

Data Science undergraduate at **IIT Madras** and Chemical Engineering student with expertise in Machine Learning, AI Architecture, and Full-Stack Engineering. Creator of 140+ software repositories, 1 Technological Patent applicant, and author of 1 Published Research Paper. Specialized in building intelligent two-stage LLM classifiers, reactive web applications, and high-performance algorithms.

EDUCATION

Indian Institute of Technology Madras (IIT Madras) — **2022 – Present**
Bachelor of Science (B.S.) in Data Science & Applications — Chennai, India

Coursework: Machine Learning, Statistical Modeling, Data Structures & Algorithms, Deep Learning, Database Systems.

Chemical Engineering Degree — **2022 – Present**
Bachelor of Technology (B.Tech) / Engineering Studies

KEY MILESTONES & CREDENTIALS

- **1 Technological Patent Filed:** Invented and submitted an official technological patent for an engineering system and computational methodology.
- **1 Published Academic Research Paper:** Authored and published research detailing algorithmic optimizations and data structures.
- **140+ Software Repositories Shipped:** Engineered full-stack web applications, AI classifiers, computer vision modules, and visual canvas tools available on GitHub.

FEATURED PROJECTS & CASE STUDIES

Two-Stage LLM Classifier & Vector Query Router | **Next.js 15, TypeScript, Supabase Vector (Pgvector), OpenAI API, Python**
- Designed a multi-stage intent classification engine routing queries to cached micro-models vs heavy LLM instances.
- Achieved a **42% reduction in median API latency** and lowered compute token costs while maintaining 99.4% intent accuracy.

Computational Predictive Analytics Engine | **Python, PyTorch, Pandas, Scikit-Learn, PCA**
- Built a high-variance dataset analytics pipeline using Principal Component Analysis and custom loss scaling functions.
- Eliminated model overfitting on complex multidimensional data, achieving benchmark research results.

Interactive In-Browser ML Neural Canvas | **HTML5 Canvas, JavaScript, Softmax Heuristics**
- Developed a zero-latency client-side digit classifier extracting spatial pixel density features and rendering probability spectrums.

Academic Brand Platform | **React, TypeScript, TailwindCSS, Vite**
- Engineered responsive promotional platforms and digital media rendering architecture for academic support platforms.

TECHNICAL SKILLS

Programming Languages:	Python, C++, TypeScript, JavaScript, SQL, C, HTML5, CSS3/Sass, Bash
Machine Learning & AI:	PyTorch, Scikit-Learn, Pandas, NumPy, OpenCV, Vector Embeddings (Pgvector), LLM Prompt Engineering, Intent Routers

Web & Cloud Frameworks:	Next.js (App Router), React 18/19, TailwindCSS, WebGL/Three.js, Canvas 2D/3D, Node.js, Express, RESTful APIs
Databases & DevOps:	Supabase, PostgreSQL, Git, GitHub Actions CI/CD, Docker, Vercel, Vite, Linux/Shell